

Conserved dynamics make the solute hyperuniform, not the domains

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Abstract

Hyperuniformity—the anomalous suppression of long-wavelength density fluctuations—is increasingly used to classify spatial patterns in physical, biological, and geological systems. A pattern is normally classified by locating its constituent objects and measuring the structure factor of the resulting point set. We show that for a system with a locally conserved order parameter this procedure can give the wrong answer, because the conserved quantity and the counted objects are not the same thing. Using Cahn–Hilliard coarsening of a two-phase alloy free energy on periodic grids up to 1024^2 , we measure the spatial order of the solute-rich domains against matched complete-spatial-random, hard-core, and perturbed-lattice nulls. The domains are locally more regular than a hard-core gas (Clark–Evans $R = 1.73$, $z = +73$, against 1.42 for random sequential adsorption). The concentration field is hyperuniform, with $S_c(k) \sim k^{3.81 \pm 0.08}$. The point pattern of domain centroids is not: $S(k \rightarrow 0)$ settles on a plateau near 0.14 and the number variance grows with window area, and this survives a doubling of the box while the bias of the estimators themselves disappears. Weighting the same points by domain solute mass restores hyperuniformity, $S(k \rightarrow 0) = 0.021$. In common windows the solute-mass variance is a factor of eight below what independent domains of the observed size distribution would produce, and the domain count and the mean domain size anticorrelate ever more strongly with window size. Number fluctuations are absorbed by compensating size fluctuations: permuting the domain masses over the same positions returns the budget exactly to its independent-domain baseline, locating the entire effect in the coupling between mass and position. Nearest-neighbour regularity and long-range order must be measured separately, and so must counts and conserved quantities.

1 Introduction

A point pattern is *hyperuniform* if its large-scale density fluctuations are anomalously suppressed: the structure factor vanishes in the long-wavelength limit, $S(k) \rightarrow 0$ as $k \rightarrow 0$, equivalently the variance of the number of points in a window of radius R grows more slowly than the window’s volume—in two dimensions like its perimeter, $\sigma^2(R) \sim R$, rather than like its area, $\sigma^2(R) \sim R^2$ [1, 2]. The concept has proved unusually portable. Disordered hyperuniform patterns have been identified in avian photoreceptor mosaics [3], in jammed packings, in active and driven matter, and the diagnostic is now routinely applied to patterns in biology and the geosciences.

Portability has a cost. Hyperuniformity is defined for a density, but it is almost always *measured* on a point pattern obtained by locating discrete objects—cells, particles, grains, depressions—and treating their centroids as the density’s carriers. The step from the density to the point pattern is rarely examined. It is exact when every object carries the same weight, and it is a genuine approximation otherwise.

Two recent results frame the question we ask here. On the measurement side, Chen [6] applied the full protocol—a local diagnostic, the structure factor, the number variance, and matched nulls—

to fields of submarine pockmarks, seabed depressions left by escaping fluid. The nearest-neighbour spacing of one field was more regular than a hard-core gas, the kind of local order that invites a hyperuniform interpretation; yet every field was long-range super-Poissonian, with $S(k \rightarrow 0)$ between 2.5 and 19.4, and the hyperuniform region of the order plane was empty. The methodological conclusion was that a local statistic cannot substitute for a long-range one. That system, however, is one in which long-range order simply does not exist, so the point-pattern step was never put under stress.

On the theory side, systems governed by a locally conserved order parameter are the natural place where long-range order *does* exist. The Cahn–Hilliard equation [7] is a continuity equation: a composition fluctuation of wavelength $2\pi/k$ can only be built by transporting solute across that distance, so long-wavelength modes are kinematically frozen while short-wavelength ones grow. The resulting suppression of the small- k structure factor of the concentration field is long established: Yeung [8] proved that a scaling coarsening pattern with conserved dynamics must have $S_c(k) \sim k^\delta$ with $\delta \geq 4$, numerical work found $\delta \simeq 4$ [9], Tomita [10] gave the scaling argument that explains it, and De Luca *et al.* [11] recently extended that argument to active and non-constant-mobility cases, again obtaining 4. We claim no novelty whatever for this, and reproduce it here only as an internal check on the simulation.

The question we ask is what happens to the *point pattern* in such a system. Mass conservation constrains the solute; it does not constrain how many domains hold it. A region can meet its solute budget with a few large domains or with many small ones, and nothing in the dynamics forbids the trade. Whether the centroids inherit the field’s hyperuniformity is therefore an empirical question, and the answer bears directly on every study that classifies a conserved system by counting objects.

That the answer can be negative is, in general, now known. Torquato *et al.* [4] have generalised hyperuniformity to particles carrying weights—charges, dipoles, Voronoi volumes—and shown that a hyperuniform arrangement need not remain hyperuniform when weighted, and that a non-hyperuniform one may become so; related weighted point constructions have been examined in an ecological setting [5]. The general statement that positions and weights are different objects is therefore established, and we do not claim it.

What this paper adds is the conserved-dynamics instance and its mechanism. First, we exhibit a system in which the weight is not an arbitrary internal degree of freedom but the conserved quantity itself, and show that the field, the centroid count pattern, and the mass-weighted centroid pattern of one simulation fall into different classes—so that the discrepancy is not a curiosity of exotic weights but the generic situation whenever discrete objects share out a conserved density. Second, we quantify the difference: a variance budget, built from an identity and three measured inputs with no free parameter, shows the observed solute rigidity to be a factor of eight beyond what independent domains of the observed size distribution would give, with the shortfall supplied by an anticorrelation between domain count and mean domain size that deepens monotonically with scale. Third, we draw the methodological consequence for the many studies that classify such systems by locating and counting objects.

2 Model and methods

2.1 The system

We integrate the Cahn–Hilliard equation

$$\frac{\partial c}{\partial t} = \nabla \cdot \left[M \nabla \left(\frac{\partial f}{\partial c} - \kappa \nabla^2 c \right) \right], \quad (1)$$

for a conserved composition $c(\mathbf{r}, t)$ on a periodic $n \times n$ grid, with a semi-implicit pseudospectral scheme. The homogeneous free energy $f(c)$ is a double well with a solute-poor and a solute-rich

minimum, calibrated in earlier work for a solid solution of a minor element in a sulphide host; the specific chemistry is immaterial here and is reported elsewhere. What matters for the present argument is only that f has two minima, that the initial composition lies inside the spinodal, and that the dynamics conserve $\int c d\mathbf{r}$ locally rather than merely globally. All runs use $c_0 = 0.30$, off-critical, so that the minority phase forms discrete domains rather than a bicontinuous structure and centroids are well defined.

The gradient coefficient κ is fixed by requiring the fastest-growing spinodal wavelength to be 12 grid cells, and the time step by requiring the fastest-growing mode to be amplified by 2% per step. The absolute length scale is therefore set by numerical resolvability and not by the material; accordingly, every quantity reported below is dimensionless—ratios to a Poisson reference, scaling exponents, and correlation coefficients—and none depends on that choice.

The principal ensemble is four independent runs at $n = 1024$, coarsened isothermally for 2000 steps, at which point each configuration contains 2736 ± 20 domains with a mean spacing of 19.6 cells. A second ensemble of six runs at $n = 512$, carried to 4000 steps, is used for the finite-size comparison of Sec. 3.4. We stop the large ensemble at the stage that maximises the accessible wavevector range rather than at the latest attainable time: coarsening reduces the domain count, and with it the statistical quality of precisely the long-wavelength band the paper is about.

2.2 From field to point pattern

Domains are the connected components of $c > c_{\text{thr}}$, with c_{thr} the midpoint between robust percentiles of the two phase compositions. Components are labelled on the torus, so that a domain straddling the periodic boundary is one domain and not two. Each domain contributes one point, the centroid of its excess composition $c - c_{\text{thr}}$, computed as a circular mean so that the wrap is handled correctly, and one weight, its solute mass $m_i = \sum (c - c_{\text{thr}})$ over the component. Components smaller than eight cells are discarded as unresolved ripples. Moving c_{thr} across the central 40% of the composition gap changes the exponents reported below by less than 0.05.

2.3 Diagnostics

Local order. The Clark–Evans index [13] $R = \bar{d}_{\text{NN}}/[1/(2\sqrt{\rho})]$, with $R < 1$ clustered, $R = 1$ random, $R > 1$ regular. On a torus the minimum-image distance is exact and no edge correction arises—an advantage over field surveys, where boundary bias is the dominant systematic.

Long-range order. The structure factor is evaluated on the exact reciprocal lattice of the periodic box,

$$S(\mathbf{k}) = \frac{1}{N} \left| \sum_{j=1}^N e^{-i\mathbf{k}\cdot\mathbf{r}_j} \right|^2, \quad \mathbf{k} = 2\pi(n_x, n_y)/L, \quad (2)$$

with no window function and hence none of the low- k leakage that complicates bounded surveys, and radially averaged in logarithmic shells. For an uncorrelated set of N points on this lattice $\langle S \rangle = 1$ for every $\mathbf{k} \neq 0$. The weighted version, $S_w(\mathbf{k}) = |\sum_j w_j e^{-i\mathbf{k}\cdot\mathbf{r}_j}|^2 / \sum_j w_j^2$, is normalised so that uncorrelated positions again give unity whatever the weights; with $w_j = m_j$ it is a spectrum of the solute rather than of the domains. The number variance $\sigma^2(R)$ is taken over disks placed at random on the torus. We fit the low- k exponent α in $S \sim k^\alpha$ between the smallest available wavevector and $0.45k_{\text{peak}}$, and quote it with its standard error.

2.4 Two constraints that must not be read as results

First, the scheme conserves the mean composition exactly and the box is periodic, so $S_c(k=0) = 0$ identically. That is arithmetic, not physics; only $k > 0$ is fitted or plotted. Second, a configuration with N points in a periodic box has a number variance driven to zero as the window approaches the whole box, for the same trivial reason. Windows are therefore restricted to $R \leq L/6$ (Sec. 2.4), and—importantly—the complete-spatial-random null is generated at *fixed* N , so that it carries the identical constraint and the comparison remains fair.

2.5 Nulls and validation

Each configuration is compared against complete spatial randomness (CSR) at matched N ; a random-sequential-adsorption (RSA) hard-core process with exclusion matched to the observed closest approach, representing steric exclusion without organisation; and a Gaussian-perturbed square lattice as a known hyperuniform reference. Passed through the identical pipeline, CSR returns $R = 0.99$, $S(k \rightarrow 0) = 1.14$, and a number-variance slope of 2.02, against the ideal 1, 1, and 2; the perturbed lattice returns $\alpha = +2.02$ and slope 0.93. The estimators are therefore doing what they are supposed to, and the null results reported below are not artefacts of the pipeline.

3 Results

Table 1: The three objects and the nulls, at 1024^2 after 2000 steps ($N = 2736$ domains per configuration, four realisations). R is the Clark–Evans index, $H = S(k \rightarrow 0)/S(k_{\text{peak}})$, and the last column the log–log slope of the number variance fitted above 1.4 domain spacings.

	R	$S(k \rightarrow 0)$	α	H	σ^2 slope
Domain centroids (counts)	1.73	0.143	−0.14(4)	0.089	1.80
Same points, mass-weighted	—	0.021	+0.40(9)	0.014	—
Concentration field	—	—	+3.81(8)	—	—
CSR (fixed N)	0.99	1.14	−0.05	—	2.02
RSA hard core	1.42	0.24	+0.24	0.20	1.79
Perturbed lattice	1.45	0.017	+2.02	0.008	0.93

3.1 The domains are locally more regular than a hard-core gas

Table 1 collects the results discussed in this section.

The centroid pattern has $R = 1.73$ ($z = +73$), well above both CSR (0.99) and the RSA hard-core null (1.42) at the same density [Fig. 1]. Local order is thus real and is not merely exclusion: the domains are spaced more evenly than hard discs of the same closest approach would be, as expected when each domain draws solute from, and so depletes, its neighbourhood.

3.2 The field is hyperuniform

The concentration field has $S_c(k) \sim k^{\alpha_c}$ with $\alpha_c = 3.81 \pm 0.08$ over more than a decade in k [Fig. 2(a)], recovering the established behaviour of conserved coarsening [8, 9, 10, 11]. We report it as a check on the simulation, not as a finding.

The fitted value lies nominally below the bound $\delta \geq 4$ of Ref. [8], by about 2.4 standard errors of the fit. We do not read this as a violation. The bound holds asymptotically for a scaling state,

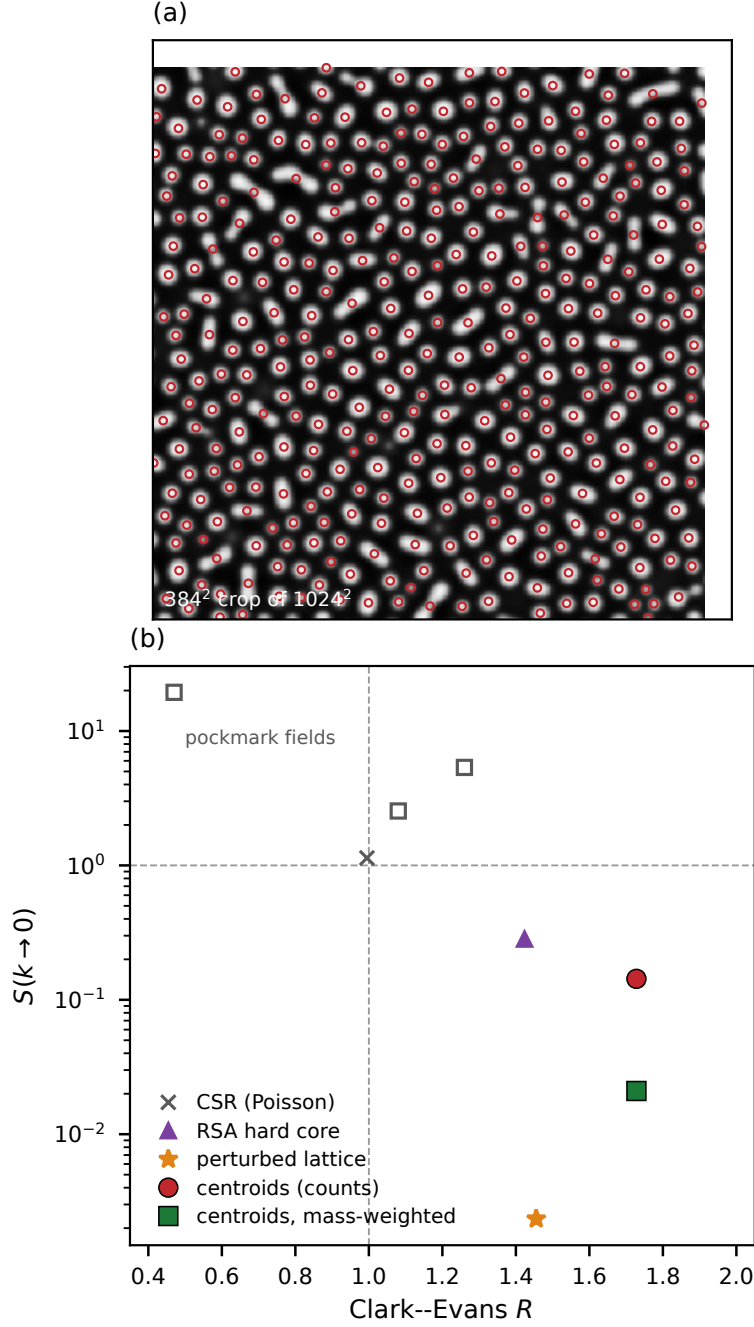


Figure 1: (a) Solute-rich domains (grey scale) and their mass-weighted centroids (circles), a 384^2 crop of one 1024^2 configuration after 2000 steps. (b) Order plane: Clark–Evans index against the long-wavelength structure factor. The domains are locally regular, exceeding the hard-core null, but their count spectrum sits well above the hyperuniform limit; weighting the same points by solute mass moves them down by nearly an order of magnitude. Grey squares are the three Mediterranean pockmark fields of Ref. [6], shown for scale comparison only.

whereas the fit is taken over a finite window that ends at $0.35k_{\text{peak}}$, where curvature towards the peak is already depressing the local slope; restricting the fit to the lowest half-decade returns a steeper exponent with a correspondingly larger error. The quoted uncertainty is that of the regression alone

and does not include this systematic.

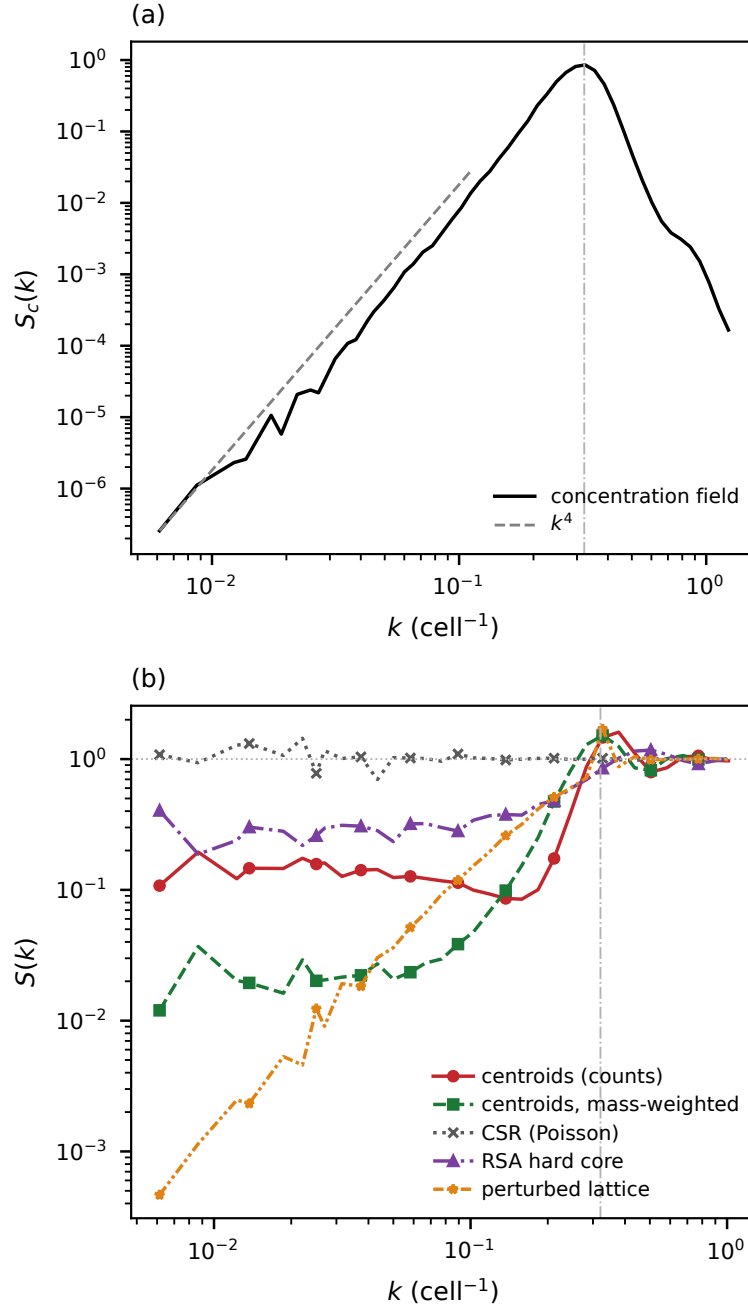


Figure 2: (a) Structure factor of the concentration field, with a k^4 guide. (b) Spectra of the point patterns and nulls. The two centroid curves are the *same points*, differing only in whether each point carries unit weight or its domain's solute mass. Every series is distinguished by line style and marker as well as colour, so the figure is legible in monochrome. The dot-dashed vertical line marks the coarsening peak k_{peak} .

3.3 The point pattern is not

The centroid pattern is not hyperuniform. Its structure factor falls well below the Poisson line but then flattens, settling on a plateau $S(k \rightarrow 0) = 0.143$ with a low- k exponent $\alpha = -0.14 \pm 0.04$ [Fig. 2(b)], and its number variance grows as $\sigma^2 \sim R^{1.80}$, close to the area-like exponent 2 and far from the perimeter-like 1 returned by the perturbed lattice [Fig. 3(a)]. By the criterion that defines the class, the pattern is not hyperuniform; by the ratio of the plateau to the peak, $H = 0.089$, it is not effectively hyperuniform either. It is, however, strongly suppressed: its number variance lies a factor of four to five below CSR at every window size, and below the hard-core null as well.

3.4 Doubling the box does not move the plateau

A plateau at the smallest accessible wavevector is exactly what a box too small to resolve the true limit would produce, so we repeated the measurement at $n = 512$ and $n = 1024$, halving $k_{\min}$ from 0.0123 to 0.0061 cell⁻¹. At the same coarsening stage the two spectra coincide over their shared range, and the additional half-decade of low k opened by the larger box shows the same plateau [Fig. 3(b)]. The exponents agree within their errors, -0.20 ± 0.08 at 512² against -0.14 ± 0.04 at 1024².

The strongest form of this argument is asymmetric. Between the two box sizes the *estimators* improve markedly—the CSR control moves from $(S, \text{slope}) = (0.92, 1.85)$ to $(1.14, 2.02)$, converging on its theoretical values as the finite-box bias is removed—while the data do not move at all. A residual bias that vanishes for the null but leaves the measurement unchanged cannot be what produces the measurement.

3.5 Weighting by solute mass restores hyperuniformity

Weighting the identical set of points by domain solute mass drops the long-wavelength plateau from 0.143 to 0.021 and turns the exponent positive, $\alpha = +0.40 \pm 0.09$, with $H = 0.014$ [Fig. 2(b)]. Nothing about the positions has changed. What has changed is which quantity is being counted, and the two answers differ by nearly an order of magnitude.

3.6 A negative result: filtering by size does not isolate the cause

The obvious way to test whether polydispersity is responsible is to restrict the analysis to domains of nearly equal mass. It does not work, and the way it fails is instructive. Selecting a subset removes points, and random removal by itself drives $S(k)$ towards the Poisson value—for independent thinning with retention p , $S_{\text{thinned}}(k) = 1 + p[S_{\text{full}}(k) - 1]$ —so the comparison must be against a random subset of the same size, not against the full pattern. Made that way, the test returns the wrong sign: retaining the 30% of domains closest to the median mass gives $S(k \rightarrow 0) = 0.963$ against 0.616 for matched random thinning, and the discrepancy grows as the selection is tightened (Table 2).

The diagnosis is that the two subsets have *identical* Clark–Evans indices (1.19 versus 1.19 at $q = 0.30$; 1.35 versus 1.35 at $q = 0.50$). The excess is therefore not local: it is long-wavelength. Domain mass is spatially autocorrelated on large scales, so a subset selected by mass is also a subset selected by position, and the selection injects exactly the kind of long-wavelength density variation the measurement is designed to detect. A hard filter cannot separate polydispersity from the spatial autocorrelation of the filtering variable. We report the result because it motivates the analysis of the next section, which deletes nothing.

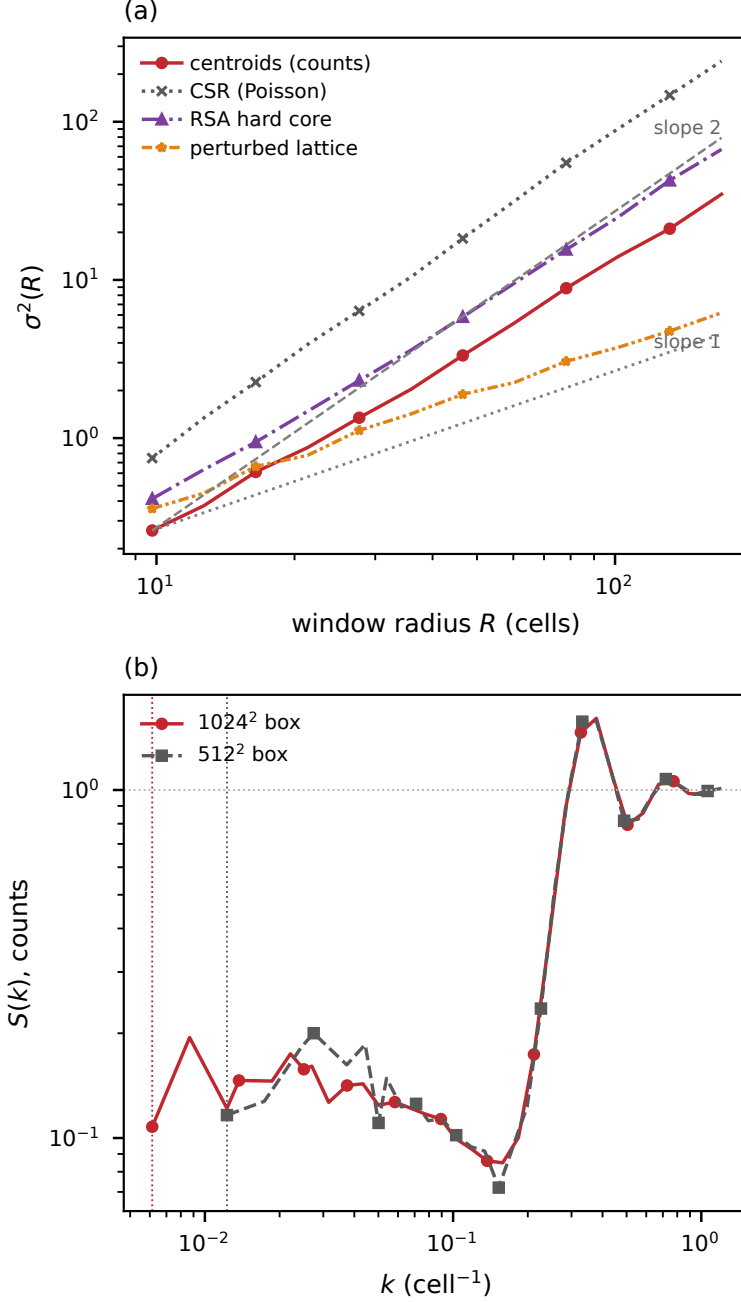


Figure 3: (a) Number variance against window radius, with slope-1 and slope-2 guides. (b) Finite-size check: the count spectrum in boxes of 512^2 and 1024^2 at the same coarsening stage; dotted verticals mark the respective $2\pi/L$. Doubling the box halves the accessible $k_{\min}$ and leaves the plateau where it was, while over the same change the CSR control moves from $(S, \text{slope}) = (0.92, 1.85)$ to $(1.14, 2.02)$. The bias that vanishes for the null is not what produces the measurement.

4 The variance budget

4.1 An identity with no free parameter

For a sum of N masses drawn independently of one another and of N , the variance of a compound random sum is

$$\frac{\text{Var}[M]}{\langle m \rangle^2} = \langle N \rangle_8 \text{CV}^2 + \text{Var}[N], \quad (3)$$

Table 2: The negative filtering test of Sec. 3.6. Retaining the fraction q of domains whose mass is closest to the median is compared with a random subset of the same size drawn from the same configuration. The filtered subsets have higher long-wavelength fluctuations, and identical local order.

q	CV	$S(k \rightarrow 0)$	$S(k \rightarrow 0)$	R	R
	kept	filtered	thinned	filtered	thinned
0.30	0.12	0.963	0.616	1.19	1.19
0.50	0.21	0.599	0.587	1.35	1.35
0.70	0.31	0.482	0.417	1.51	1.51

with CV the coefficient of variation of the domain masses. Every quantity on the right is measured: $\langle N \rangle$ and $\text{Var}[N]$ in the same windows in which M is accumulated, and $\text{CV} = 0.528$ from the mass distribution. Equation (3) is therefore a prediction with no adjustable parameter, and the residual

$$C \equiv \frac{\text{Var}[M]}{\langle m \rangle^2} - \langle N \rangle \text{CV}^2 - \text{Var}[N] \quad (4)$$

is the part of the observed solute rigidity that independence cannot account for.

The value of this construction is that CV is an input that the window statistics do not already contain. It converts the qualitative statement “size compensates count” into a number: how much fluctuation independent domains of the observed size distribution *would* have produced.

4.2 Independence is excluded by a factor of eight

Table 3 gives the budget across window sizes and Fig. 4 displays it. At the widest window, $R = 170.7$ cells or about eight domain spacings, independent domains would give a variance of 100.3 in units of one domain, of which 66.6 comes from the size term and 33.8 from the count term. The measured value is 12.2. The residual is $C = -88.1$: the system cancels 88% of the fluctuation that polydispersity alone would impose.

The cancellation is not a coincidence of one window size. The ratio of measured to baseline falls monotonically from 0.38 at $R \approx 1.4$ spacings to 0.12 at $R \approx 8$ spacings [Fig. 4(b)]: the wider the window, the more completely the compensation operates. Over the same range the correlation between the domain count and the mean domain mass within the window deepens monotonically from -0.62 to -0.89 , while the spread of the window-mean domain mass contracts from 0.221 to 0.031. Where there are more domains, the domains are smaller.

Table 3: The variance budget, Eq. (3), in units of one domain. $\text{CV}^2 = 0.279$ throughout. Windows narrower than 1.4 domain spacings are omitted; there the decomposition is dominated by discreteness. The last column is the correlation between the domain count and the mean domain mass in the same window.

R	$\langle N \rangle$	$\text{Var}[N]$	$\langle N \rangle \text{CV}^2$	baseline	$\text{Var}[M]/\langle m \rangle^2$	$\text{corr}(N, \bar{m})$
27.7	6.3	1.36	1.75	3.11	1.17	-0.62
35.9	10.6	2.14	2.95	5.09	1.57	-0.68
46.5	17.7	3.38	4.95	8.33	2.19	-0.74
60.4	29.9	5.50	8.33	13.83	3.03	-0.79
78.3	50.2	8.49	14.00	22.49	4.14	-0.83
101.5	84.4	13.75	23.56	37.31	5.95	-0.86
131.6	141.8	22.81	39.57	62.38	8.27	-0.88
170.7	238.6	33.76	66.57	100.33	12.24	-0.89

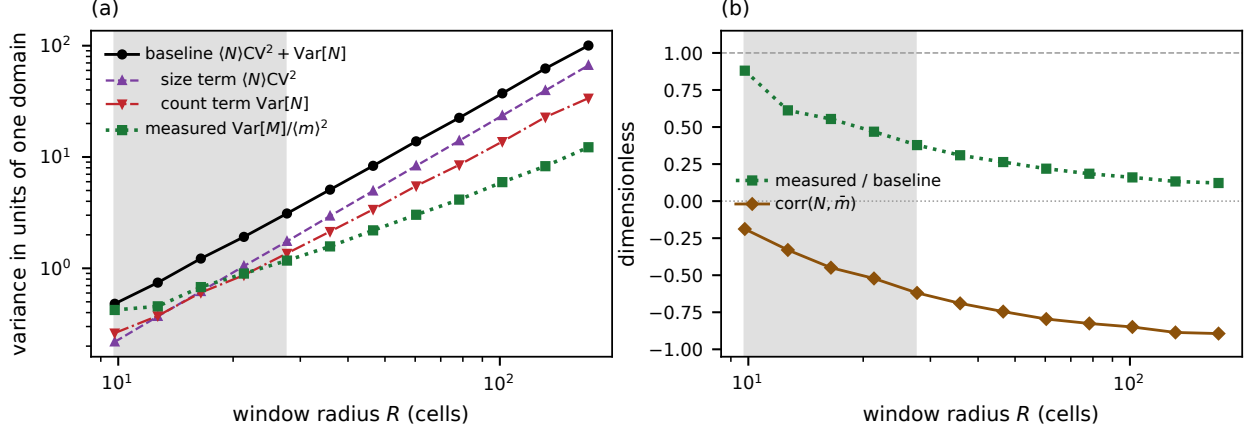


Figure 4: The variance budget, Eq. (3), in units of one domain. (a) What independent domains would produce, split into its size and count terms, against the measured solute-mass variance. The single input not already contained in the window statistics is the coefficient of variation of the domain masses, $CV = 0.528$; it is what makes the baseline a prediction with no free parameter. (b) The surviving fraction of that baseline, and the correlation between the domain count and the mean domain mass in the same window. *These two curves are not independent evidence*: where the solute content of a window is nearly rigid, $\bar{m} \propto 1/N$ is close to an identity and $\text{corr}(N, \bar{m}) \rightarrow -1$ follows algebraically. The correlation is shown because it locates the scale on which the compensation sets in, not as a second measurement. Shading marks windows narrower than 1.4 domain spacings, where the decomposition is dominated by discreteness.

4.3 What the budget does and does not establish

Three limitations bound this analysis, and we state them explicitly because each could otherwise be over-read.

First, the correlation $\text{corr}(N, \bar{m}) \rightarrow -0.89$ and the smallness of $\text{Var}[M]$ are *not* independent pieces of evidence. If the solute content of a window is nearly rigid, then $\bar{m} \propto 1/N$ is close to an identity and a correlation approaching -1 follows algebraically. The correlation is reported because it expresses the same fact in the language of the mechanism and supplies the scale on which the compensation sets in, not because it constitutes a second measurement.

Second, the residual C mixes two effects: correlation between a domain's mass and the local domain count, and correlation between the masses of neighbouring domains. The decomposition itself does not separate them; the measurements of Sec. 4.4 do.

Third, the budget should not be inverted to predict the plateau height of Sec. III C. Doing so would require the additional assumption that domain positions are determined by the mass field, which we have not established. The real-space cancellation and the reciprocal-space plateau are two views of one physical situation and are reported side by side, not derived from one another.

4.4 Attribution by mark permutation and mass autocorrelation

Two further measurements close the budget. Both treat the domain masses as marks attached to the observed positions, and both use a mark permutation as their null: the masses are randomly reassigned among the same domains, which destroys any relation between mass and position while leaving the positions, the mass distribution, and hence CV exactly as observed.

The baseline is not an approximation. Recomputing the budget with permuted marks returns residuals consistent with zero at every window size, and a ratio of measured to baseline variance of 0.99–1.09 against 0.88–0.12 for the real marks. Equation (3) therefore describes these configurations

exactly once the marks are made independent of position, which is what it assumes; the entire residual $C = -88$ is attributable to mass–position coupling and none of it to a failure of the identity.

The coupling is with crowding, not with other masses. The mark covariance $c_{mm}(r) = \langle \delta m_i \delta m_j \rangle_r / \sigma_m^2$ departs from its permutation envelope only out to about two domain spacings and lies inside it, indistinguishable from zero, from ≈ 50 cells to the largest separation measured, 232 cells [Fig. 5(a)]. Domain mass carries no large-scale spatial structure. The correlation between a domain’s mass and the number of neighbours within r , by contrast, reaches -0.68 at one domain spacing and decays slowly to -0.08 at 128 cells [Fig. 5(b)], against a permuted value of $+0.01$ throughout. Crowded domains are smaller. The compensation that makes the solute rigid is therefore local and steric in origin rather than a large-scale organisation of domain sizes.

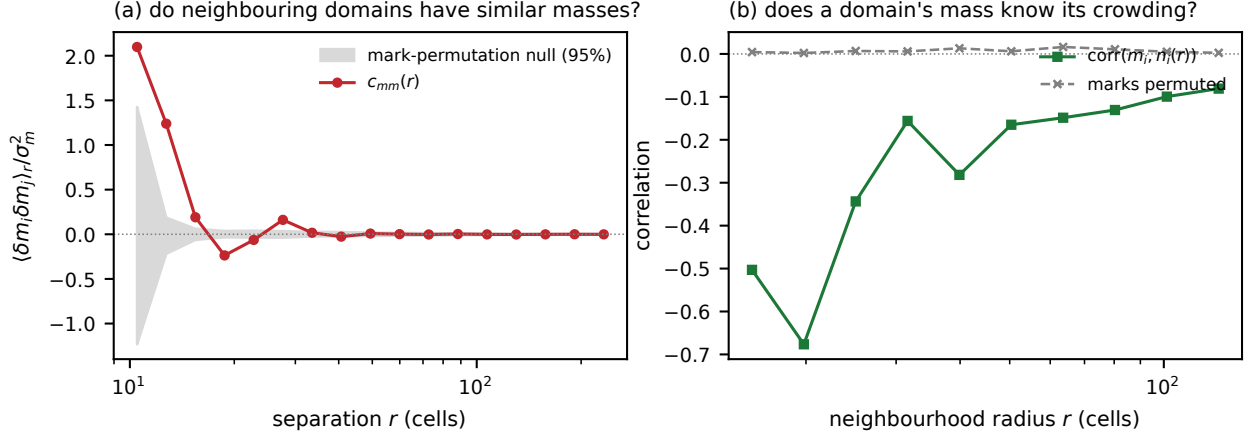


Figure 5: Domain mass treated as a mark on the point pattern, against a mark-permutation null in which the masses are reassigned at random among the same domains. (a) Mark covariance. The signal is confined to separations below about two domain spacings; beyond ≈ 50 cells it is inside the null envelope out to 232 cells, so domain mass has no large-scale spatial structure. Values exceeding unity in the first two bins reflect the small number of pairs available there, where the null envelope is correspondingly wide, and should not be interpreted. (b) Correlation between a domain’s mass and the number of neighbours within r : crowded domains are smaller. This is the coupling that supplies the residual of the variance budget.

5 Discussion

5.1 Relation to weighted hyperuniformity

The formal apparatus for what we measure here exists. Reference [4] defines weighted spectral densities and local variances for particles carrying scalar or tensorial weights, and establishes that the weighted and unweighted classifications are logically independent. Our contribution is not that apparatus but a physical instance of it in which the weight is fixed by a conservation law rather than chosen: the solute mass of a coarsening domain. That case has, we believe, a status the general framework does not confer on it. When the weight is a conserved density, the weighted spectrum is not one of many possible characterisations—it is the one the dynamics constrain, and the unweighted spectrum is the one experiment usually reports. The gap between them is then not a mathematical possibility but a systematic error with a known sign.

5.2 A rule for classifying conserved systems

The results support a short procedural rule for anyone applying hyperuniformity diagnostics to a pattern of discrete objects. (i) Measure local and long-range order separately; a high nearest-neighbour index is compatible with area-like number variance, as it is here and as it was in the pockmark fields [6]. (ii) If the system has a conserved quantity, measure both the count spectrum and the spectrum weighted by that quantity; here they differ by a factor of seven and fall on opposite sides of the classification boundary. (iii) Run the null through the identical pipeline and report what it returns, because the credibility of a negative result rests entirely on the estimator having been shown to work.

The rule applies well beyond the present model. Precipitate populations, droplet emulsions, cluster distributions in reaction-diffusion systems, and ecological patch mosaics are all conserved or quasi-conserved systems that are routinely characterised by locating objects and counting them.

5.3 The empirical case

Domain point patterns in a phase-separating system have been measured experimentally. In phase-separated DNA droplets, Wilken *et al.* [12] report structure factors and number variances whose long-wavelength fluctuations do not vanish, and attribute the failure to incomplete equilibration and to kinetic arrest of the coalescence process. The present results suggest that a further contribution is available without invoking either: even in a fully passive, unarrested, isothermally coarsening system, the count spectrum does not become hyperuniform, because the count is not the conserved variable. Where droplet volumes are measurable alongside positions, the weighted spectrum is the discriminating measurement, and our prediction is definite—it should lie well below the unweighted one.

5.4 Two scales, one lesson

The pockmark analysis of Ref. [6] and the present work sit at opposite ends of a very wide range of scales—seabed depressions tens of metres apart, and nanometre-scale precipitates—and reach the same methodological conclusion from opposite directions. There, local regularity exceeding a hard-core null coexisted with strongly super-Poissonian long-range behaviour, and the warning was that local order must not be read as long-range order. Here, long-range order genuinely exists, and yet the standard point-pattern measurement misses it, because the conserved quantity is not the counted object.

The two systems also differ in a way worth stating. In the pockmark fields the excess long-wavelength fluctuation was traced to substrate heterogeneity: the inhomogeneous-Poisson nulls reproduced the observed $S(k \rightarrow 0)$, so the long-range signal was a property of the geological background rather than of the pattern-forming process. In the simulations there is no substrate, no heterogeneity, and no external field, and the residual count fluctuation is generated by the pattern-forming dynamics themselves. The lesson common to both is that the long-range behaviour of a point pattern is governed by something other than the mechanism that sets its local spacing.

5.5 An unexplained feature

The low- k exponent of the count spectrum is small but significantly negative, $\alpha = -0.14 \pm 0.04$: the plateau turns slightly upward at the smallest accessible wavevectors. We have no explanation, and we have now excluded the three candidates we could think of.

It is not a finite-size artefact: Sec. 3.4 shows the upturn surviving a doubling of the box, over which the bias of the estimators themselves demonstrably disappears. It is not a thresholding artefact: the

exponent moves by less than 0.05 as the phase cut is swept across the central 40% of the composition gap. And it is not a large-scale modulation of domain size—the explanation we regarded as most likely before measuring it. Were the field to contain regions of systematically larger domains, those regions would hold fewer of them, and a long-wavelength count fluctuation would follow; but the mark covariance of Sec. 4.4 shows no mass correlation whatever beyond two domain spacings, out to a separation of 232 cells, while the excess in $S(k)$ appears at wavelengths several times longer than that.

What remains is a small excess of long-wavelength count fluctuation in a pattern whose local order is strong, whose solute is rigid, and whose domain sizes are spatially uncorrelated on those scales. We report it as an open question rather than assign it to the nearest available correlate.

5.6 Limitations

The analysis is two-dimensional, isothermal, at a single composition, and uses a single free-energy form. The domain count per configuration is $\mathcal{O}(10^3)$ and the ensemble is four realisations, so the accessible wavevector range spans about 1.7 decades; we therefore report exponents with their uncertainties over that window and do not assign a hyperuniformity class. No experimental point pattern is analysed here, and the natural test of the prediction—that a real precipitate population should show a count spectrum well above its mass-weighted spectrum—remains to be done on electron-microscope maps.

6 Conclusion

In a phase-separating system with a locally conserved order parameter, the solute-rich domains are locally more regular than a hard-core gas; the concentration field is hyperuniform; and the point pattern of domain centroids is not, on both the structure-factor and number-variance criteria, in a measurement whose estimators are validated in the same box and whose verdict is unchanged when the box is doubled. Weighting the same points by their solute mass recovers the field’s behaviour. A variance budget built on a parameter-free identity shows the solute-mass fluctuation to be a factor of eight below what independent domains of the observed size distribution would produce, with the shortfall supplied by a count–size anticorrelation that deepens monotonically with scale.

What is conserved is not what is counted. Before classifying a conserved system as hyperuniform or otherwise from the positions of its objects, one should ask whether the objects carry the conserved quantity in equal shares, and measure the weighted spectrum when they do not.

Data availability

The analysis code, the derived per-field statistics, and the figure scripts are openly available at <https://github.com/Ruqing1963/conserved-domain-hyperuniformity>. In addition to the summary statistics we release `domain_marks_stage2000.csv`, the complete per-domain table for the 1024^2 ensemble analysed here: 10 942 domains across four realisations, each with its realisation index, centroid coordinates in grid units, area in cells, and solute mass. Every quantity in Sec. 4.4 and in Table 3 can be recomputed from that file alone, without repeating the simulations.

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